Zero-Trust IoT Network: Baseline Setup Instructions

These instructions let someone rebuild and test the baseline network from scratch in Cisco Packet Tracer.

1. Purpose

The baseline network shows that default-deny ACLs and VLAN segmentation allow registered IoT devices to reach only their approved server, while unknown devices and unauthorized destinations are blocked.

2. Requirements

Item	Detail
Cisco Packet Tracer	Version used: (fill in)
Later phases	Python 3, Wireshark

3. Files

File	Description
`ZeroTrust_IoT_BASELINE.pkt`	Unchanged baseline. Do not edit; work in a copy.
`ZeroTrust_IoT_noACL.pkt`	Optional "before Zero-Trust" checkpoint
`R1_running-config.txt`	Router configuration
`SW1_running-config.txt`	Switch configuration
`screenshots/`	Diagram, test results, ACL counters

4. Network Design

Zones and VLANs

Zone	VLAN	Subnet	Gateway
IoT	10	192.168.10.0/24	192.168.10.1
Security	20	192.168.20.0/24	192.168.20.1
Servers	30	192.168.30.0/24	192.168.30.1
Quarantine	99	192.168.99.0/24	192.168.99.1

Devices and addressing

Device	Type	IP	Connection	VLAN
R1-Gateway	Router 2911	.1 on each subnet	Gig0/0 trunk to SW1 Fa0/1	n/a
SW1	Switch 2960	n/a	n/a	n/a
AP1	Access Point-PT	n/a	Wired to SW1 Fa0/2	10
Motion-Sensor	IoT sensor	192.168.10.11	Wireless to AP1	10
Smart-Camera	IoT webcam	192.168.10.12	Wired to SW1	10
Thermostat	IoT	192.168.10.13	Wireless to AP1	10
Door-Lock	IoT	192.168.10.14	Wireless to AP1	10
Rogue-Device	Laptop (unregistered)	192.168.10.50	Wired to SW1	10
ZT-Controller	PC	192.168.20.10	Wired to SW1 Fa0/7	20
IoT-Server	Server	192.168.30.10	Wired to SW1 Fa0/8	30
Video-Server	Server	192.168.30.20	Wired to SW1 Fa0/9	30

All IoT, PC and server addresses use mask 255.255.255.0 and are set statically (no DHCP). Wireless IoT devices use the same SSID and security settings as AP1. Confirm each wired device's switch port with `show interfaces status`, since ports were assigned in the order the cables were added.

Device registry (approved communication)

Device	Registered	Allowed destination	Allowed service
Motion-Sensor	Yes	IoT-Server 192.168.30.10	TCP 80
Smart-Camera	Yes	Video-Server 192.168.30.20	TCP 80
Thermostat	Yes	IoT-Server 192.168.30.10	TCP 80
Door-Lock	Yes	IoT-Server 192.168.30.10	TCP 80
Rogue-Device	**No**	None	None

5. Configuration

Switch (SW1)

```
enable
configure terminal
hostname SW1
vlan 10
 name IOT
vlan 20
 name SECURITY
vlan 30
 name SERVERS
vlan 99
 name QUARANTINE
exit
interface fa0/1
 switchport mode trunk
interface range fa0/2-6
 switchport mode access
 switchport access vlan 10
interface fa0/7
```

```
 switchport mode access
 switchport access vlan 20
interface range fa0/8-9
 switchport mode access
 switchport access vlan 30
end
write memory
```

Router (R1-Gateway)

```
enable
configure terminal
hostname R1-Gateway
interface gig0/0
 no shutdown
interface gig0/0.10
 encapsulation dot1Q 10
 ip address 192.168.10.1 255.255.255.0
interface gig0/0.20
 encapsulation dot1Q 20
 ip address 192.168.20.1 255.255.255.0
interface gig0/0.30
 encapsulation dot1Q 30
 ip address 192.168.30.1 255.255.255.0
interface gig0/0.99
 encapsulation dot1Q 99
 ip address 192.168.99.1 255.255.255.0
exit
ip access-list extended IOT-POLICY
 permit tcp host 192.168.10.11 host 192.168.30.10 eq 80
 permit tcp host 192.168.10.12 host 192.168.30.20 eq 80
 permit tcp host 192.168.10.13 host 192.168.30.10 eq 80
 permit tcp host 192.168.10.14 host 192.168.30.10 eq 80
 permit icmp 192.168.10.0 0.0.0.255 host 192.168.10.1
 deny ip any any
exit
interface gig0/0.10
 ip access-group IOT-POLICY in
exit
ip access-list extended QUARANTINE-POLICY
 deny ip any any
exit
```

```
interface gig0/0.99
 ip access-group QUARANTINE-POLICY in
end
write memory
```

Packet Tracer does not accept the `log` keyword on ACL lines, so it is omitted.

6. Opening and Verifying the Baseline

1. Open `ZeroTrust_IoT_BASELINE.pkt` (or follow section 5 to rebuild it).
2. Confirm all links are green and the wireless devices show dotted lines to AP1.
3. On the router, run `show ip interface brief`. Gig0/0 and all four subinterfaces should be up/up.
4. On the switch, run `show vlan brief` and `show interfaces trunk`. Fa0/1 should be trunking.
5. On the router, run `show access-lists` and `show ip interface gig0/0.10`. The second should show `Inbound access list is IOT-POLICY`.
6. Run `clear access-list counters` before testing.

Note: the IoT devices have no command line. Test traffic from them is created with **Complex PDU** (TCP, source port 1025, destination port 80, One Shot).

7. Test Procedure

After each test, record the PDU result and run `show access-lists` to see which line's counter moved.

ID	Source	Destination	Service	Expected	Line
A1	Motion-Sensor	IoT-Server	TCP 80	Success	10
A2	Smart-Camera	Video-Server	TCP 80	Success	20
A3	Thermostat	IoT-Server	TCP 80	Success	30
A4	Door-Lock	IoT-Server	TCP 80	Success	40
B1	Rogue-Device	IoT-Server	Web browser	Blocked	60
B2	Rogue-Device	IoT-Server	Ping	Blocked	60
B3	Rogue-Device	Video-Server	Ping	Blocked	60
B4	Rogue-Device	Video-Server	Web browser	Blocked	60

ID	Source	Destination	Service	Expected	Line
C1	Motion-Sensor	Video-Server	TCP 80	Blocked	60
C2	Motion-Sensor	IoT-Server	Ping	Blocked	60
C3	Smart-Camera	IoT-Server	TCP 80	Blocked	60
C4	Thermostat	Video-Server	TCP 80	Blocked	60
D1	ZT-Controller	IoT-Server	Ping	Success (control)	none
D2	Rogue-Device	Gateway 192.168.10.1	Ping	Success (known gap)	50
D3	Rogue-Device	Smart-Camera	Ping	Success (known gap)	none

Tests A1 to A4 show legitimate communication succeeding. Test C1 is the main unauthorized-destination scenario. The counters count packets, not tests, so use the PDU results for measurements.

8. Known Limitations

- Packet Tracer does not support ACL logging, so blocked traffic is dropped but not recorded.
- IoT devices have no command line and cannot change their traffic pattern, so Scenario 3 (normal to abnormal behavior) will be generated with Python in a later phase.
- Line 50 permits ICMP from the whole IoT subnet to the gateway, so the unregistered device can ping the router.
- Traffic between devices in VLAN 10 does not pass through the router, so the unregistered device can reach other IoT devices (lateral movement). Port security or per-port ACLs could address this.
- Trust scoring, logging and automated blocking are planned for the Python phase and are not part of this baseline.